



IOT-BASED SMART BATTERY MANAGEMENT SYSTEM USING ESP32 &BLYNK

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ABSTRACT

The increasing demand for efficient and reliable energy storage systems has created the need for intelligent battery monitoring and protection solutions. This project presents the design and development of an IoT-Based Smart Battery Management System using ESP32 and the Blynk IoT platform. The proposed system monitors a four-cell battery pack by measuring individual cell voltages, calculating pack voltage, average voltage, and cell imbalance conditions.

The system consists of an ESP32 microcontroller, simulated battery cells using the Wokwi platform, an LCD display for local monitoring, and protection components such as a relay and buzzer. The ESP32 continuously collects battery parameters, analyzes the battery health condition, and detects abnormal situations such as over-voltage, under-voltage, sensor faults, and cell imbalance. Based on the detected conditions, the protection system activates safety measures by disconnecting the load and generating alerts.

The collected battery data is transmitted to the Blynk IoT cloud platform through WiFi communication, enabling real-time monitoring through an intelligent dashboard. The dashboard provides live voltage data, battery health status, fault indications, diagnostics, and operational recommendations. The system uses a non-blocking programming approach with efficient state management to ensure reliable operation even during network failures.

INTRODUCTION

Battery energy storage systems are becoming increasingly important due to the rapid growth of electric vehicles, renewable energy systems, and portable electronic applications. With the increasing dependency on batteries, maintaining battery safety, reliability, and performance has become a major challenge. A Battery Management System (BMS) plays an essential role in monitoring battery parameters, protecting the battery from abnormal conditions, and improving overall battery life.

The proposed project presents an IoT-Based Smart Battery Management System using ESP32 and Blynk Cloud. The system is designed to monitor a four-cell battery pack by measuring individual cell voltages, calculating pack voltage, average voltage, and detecting cell imbalance conditions. The system consists of an ESP32 microcontroller, Wokwi-based battery cell simulation, LCD display, relay module, buzzer, and Blynk IoT dashboard for cloud-based monitoring.

The ESP32 collects real-time voltage data from each battery cell and processes the information to evaluate battery health. The system identifies various abnormal conditions such as over-voltage, under-voltage, cell imbalance, and sensor faults. During critical conditions, the protection mechanism activates by controlling the relay and buzzer to prevent possible battery damage. The LCD provides local status information, while the Blynk dashboard enables remote monitoring of battery parameters, fault alerts, diagnostics, and operational recommendations.

The proposed system focuses on developing an efficient, reliable, and intelligent battery monitoring solution suitable for applications such as electric vehicles, renewable energy storage systems, and portable power systems. By integrating IoT communication with battery management, the system improves safety, accessibility, and real-time decision-making capabilities.

Objectives

The main objectives of this project are:

- * To design and develop an IoT-based Smart Battery Management System using ESP32 and Blynk Cloud.
- * To monitor individual cell voltages of a four-cell battery pack and calculate important battery parameters such as pack voltage, average voltage, and voltage imbalance.
- * To implement battery health analysis and detect abnormal conditions including over-voltage, under-voltage, cell imbalance, and sensor faults.
- * To develop a safety protection mechanism using relay and buzzer control during critical battery conditions.
- * To provide real-time battery monitoring and remote access through the Blynk IoT dashboard.
- * To implement efficient cloud telemetry and fault monitoring using WiFi communication.
- * To develop an intelligent dashboard that displays battery status, diagnostics, alerts, and operational recommendations.

LITERATURE REVIEW

- A. Zhang et al. (2021) investigated the development of intelligent Battery Management Systems for lithium-ion battery applications. The study focused on real-time monitoring of battery parameters such as voltage, current, temperature, and State of Charge (SOC). The authors highlighted the importance of accurate battery monitoring and protection techniques to improve battery safety, reliability, and lifespan. The research demonstrated that advanced BMS solutions can effectively prevent abnormal operating conditions and enhance energy storage performance.
- B. Kumar et al. (2022) presented an IoT-based battery monitoring system using microcontrollers and cloud platforms for remote battery supervision. The study focused on transmitting battery parameters through wireless communication and displaying real-time information through a user-friendly dashboard. The authors emphasized that IoT integration enables remote diagnostics, fault detection, and improved battery management capabilities.
- C. Patel et al. (2021) proposed a smart Battery Management System for multi-cell lithium-ion battery packs. The research focused on cell voltage monitoring, voltage balancing, and protection against over-voltage and under-voltage conditions. The study showed that individual cell monitoring is essential to maintain battery pack stability and prevent cell degradation.
- D. Verma et al. (2023) investigated cloud-based battery monitoring and intelligent diagnostic systems. The proposed approach used IoT communication to collect battery data and provide real-time analysis. The authors discussed the
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advantages of cloud dashboards in monitoring battery health, identifying faults, and improving maintenance decisions.

E. Singh et al. (2024) developed an intelligent battery protection system using embedded controllers and automated fault handling techniques. The study focused on detecting abnormal battery conditions such as over-voltage, under-voltage, and imbalance conditions. The research demonstrated that integrating protection mechanisms with monitoring systems improves battery safety and operational reliability.

F. Kiran Kumar et al. (2024) proposed an IoT-enabled Battery Management System with real-time analytics and remote monitoring features. The study emphasized the use of wireless communication, cloud platforms, and intelligent dashboards for improving battery performance evaluation. The authors concluded that IoT-based BMS solutions provide scalable and efficient approaches for modern energy storage applications.

System Requirements

Hardware Components

- ESP32 Microcontroller

Main processing unit

Reads cell voltage data

Controls relay, buzzer, LCD

Communicates with Blynk through WiFi

- Battery Cell Simulation (4 Cells)

Four-cell battery pack simulation in Wokwi

Used for voltage monitoring and imbalance detection

- LCD I2C Display

Displays cell voltage, battery health, and fault status locally

- Relay Module

Provides protection by disconnecting the load during faults

- Buzzer

Provides audible alert during abnormal conditions

Software Components

- Arduino IDE

Used for ESP32 programming

- Wokwi Simulator

Used for virtual hardware simulation and testing

- Blynk IoT Platform

Used for cloud communication and dashboard monitoring

- Embedded C/C++ Programming

Used for implementing monitoring, protection logic, and IoT communication

HARDWARE DESCRIPTION

ESP32

ESP32 is the main controller used in the proposed Smart Battery Management System. It is a powerful microcontroller with built-in WiFi capability, making it suitable for IoT-based applications. The ESP32 collects voltage readings from individual battery cells, processes the data, performs battery health analysis, and communicates with the Blynk IoT cloud platform. It also controls external components such as the LCD display, relay module, and buzzer according to the system conditions.



ESP 32 Microcontroller

Four-Cell Battery Pack Simulation

A four-cell battery pack is simulated using the Wokwi platform to represent a lithium-ion battery system. Each cell voltage is monitored individually to detect voltage variations and imbalance conditions. The system calculates important parameters such as cell voltage, pack voltage, average voltage, and voltage imbalance percentage to evaluate battery performance. A four-cell battery pack is simulated using the Wokwi platform to represent a lithium-ion battery system.

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Four-Cell Battery

LCD I2C Display

An I2C-based 16×2 LCD display is used for local monitoring of battery parameters. It displays individual cell voltages, pack voltage, battery health status, and fault conditions. The LCD provides real-time information to the user without requiring cloud access.



LCD I2C Display

Relay Module

A relay module is used as a safety protection component in the battery management system. During critical conditions such as over-voltage, under-voltage, severe imbalance, or sensor faults, the relay disconnects the load to prevent possible battery damage.



Relay Module

Buzzer

A buzzer is integrated into the system to provide an audible warning during fault conditions. When a battery abnormality is detected, the buzzer is activated along with the protection mechanism to alert the user.



Buzzer

METHODOLOGY

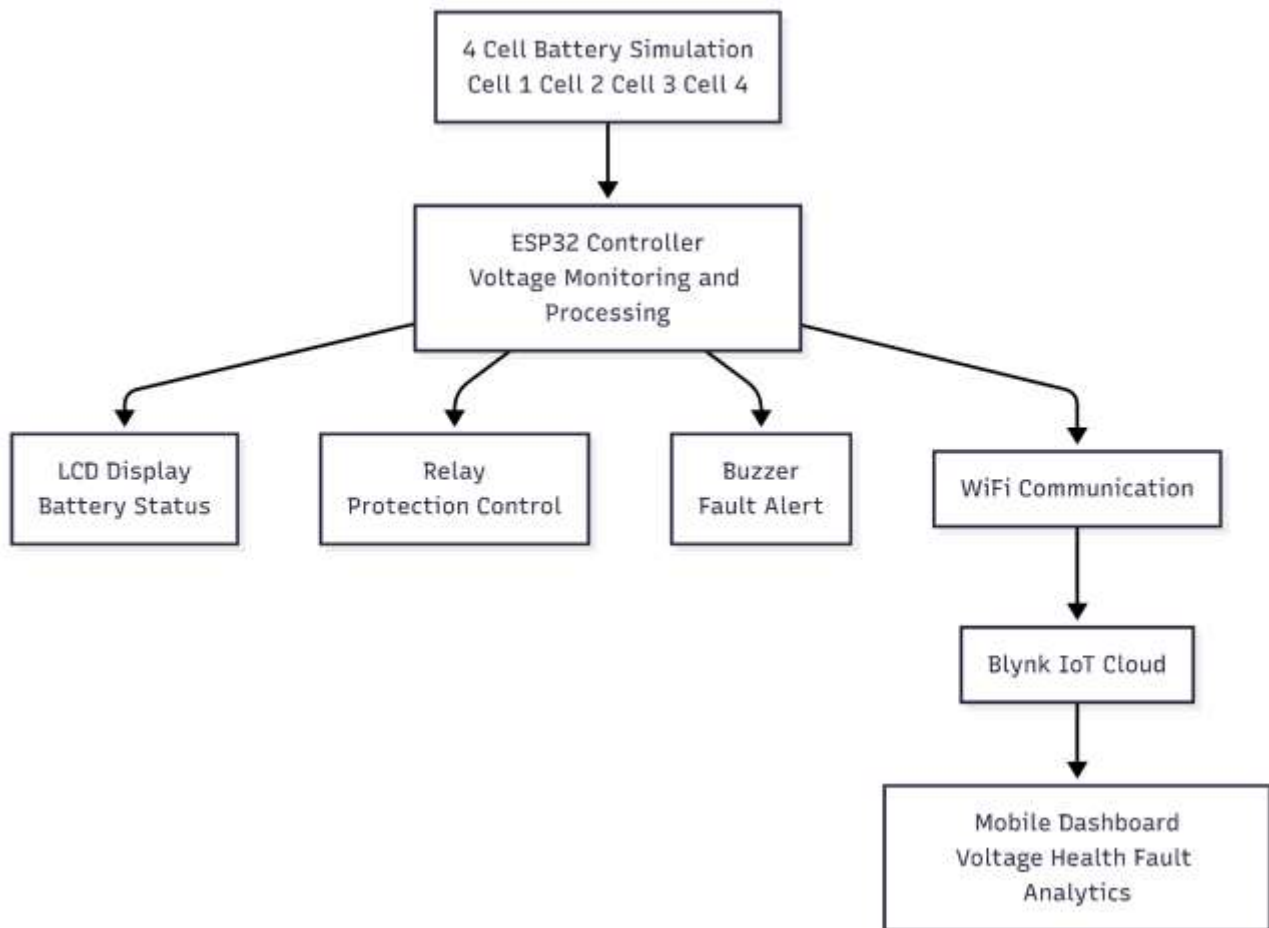
The methodology of the proposed IoT-Based Smart Battery Management System involves battery monitoring, data processing, fault detection, protection control, and cloud-based visualization. Initially, a four-cell battery pack is simulated using the Wokwi platform, where individual cell voltages are measured using the ESP32 microcontroller. The ESP32 collects voltage data from each cell and processes the readings to calculate important battery parameters such as pack voltage, average cell voltage, and voltage imbalance percentage.

The collected battery data is analyzed using predefined safety conditions to identify abnormal situations such as over-voltage, under-voltage, cell imbalance, and sensor faults. When a critical condition is detected, the protection mechanism is activated by controlling the relay and buzzer to disconnect the load and alert the user. The LCD display provides real-time local information about cell voltages, battery health, and fault status.

The ESP32 communicates with the Blynk IoT cloud platform through WiFi connectivity to transmit battery information to the mobile dashboard. The dashboard displays live battery parameters, health status, fault alerts, system mode, and operational recommendations. Event-based cloud communication is implemented to improve efficiency and ensure reliable monitoring.

The complete system is developed and tested under different battery conditions by varying cell voltages to verify normal operation, imbalance detection, and fault protection performance. The results demonstrate the effectiveness of the proposed system in providing intelligent, reliable, and real-time battery management.

.BLOCK DIAGRAM



System Architecture

Working Principle and Operation

The proposed IoT-Based Smart Battery Management System is designed to monitor, analyze, and protect a multi-cell battery pack using ESP32 and the Blynk IoT platform. The system consists of a four-cell battery simulation, ESP32 controller, LCD display, relay module, buzzer, and cloud-based monitoring dashboard. These components work together to collect battery data, identify abnormal conditions, and provide intelligent battery management.

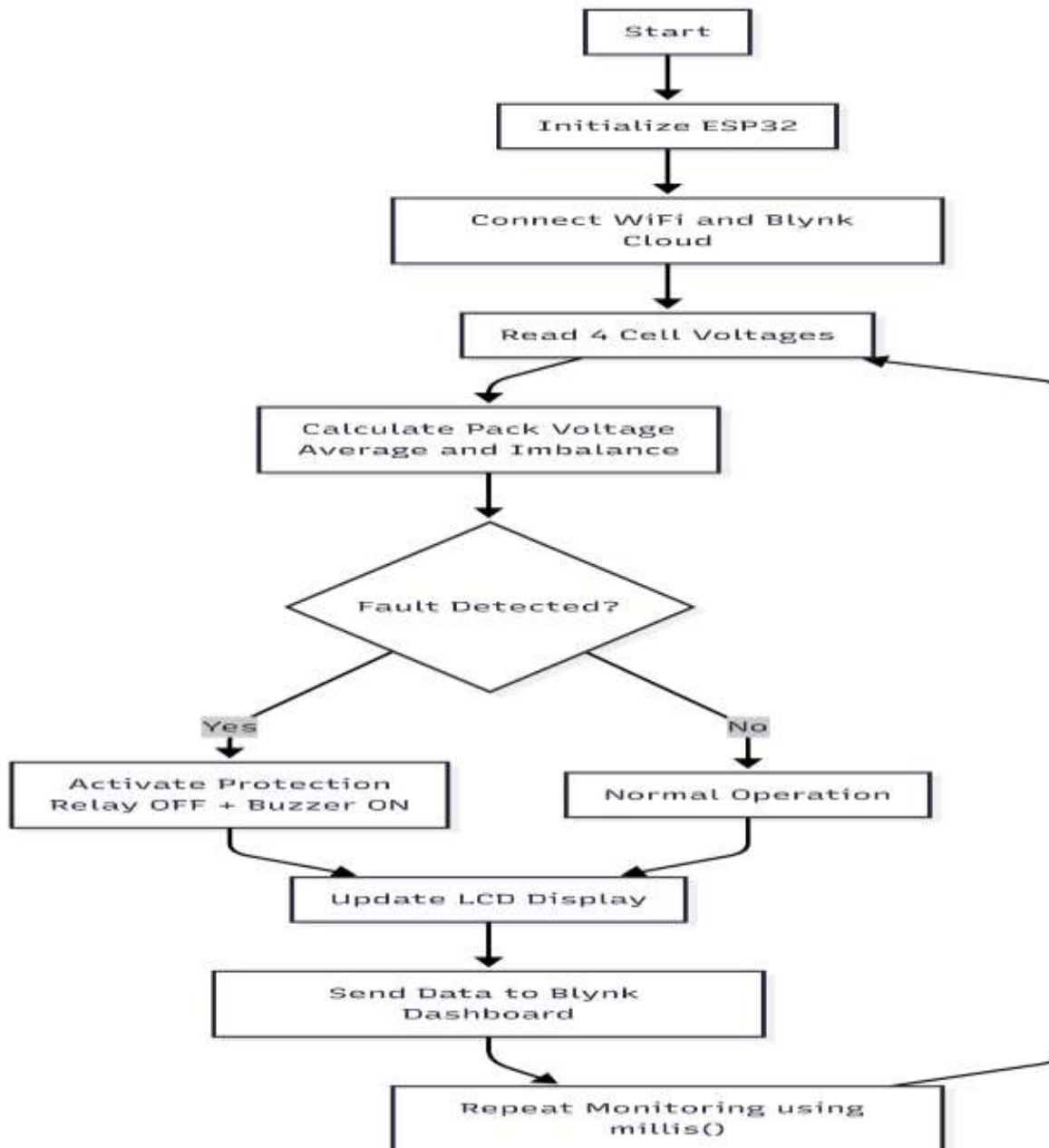
The operation of the system begins with the measurement of individual cell voltages from the four-cell battery pack. The ESP32 reads the voltage values from each cell and processes the collected data to calculate important parameters such as pack voltage, average voltage, and cell imbalance percentage. These parameters are used to evaluate the overall battery condition and health status.

The processed data is analyzed using safety monitoring algorithms. The system detects conditions such as over-voltage, under-voltage, excessive cell imbalance, and sensor faults. When an abnormal condition is identified, the protection mechanism is activated by switching OFF the relay and turning ON the buzzer to prevent possible battery damage. The LCD display provides local information about battery voltage, health status, and fault conditions.

The ESP32 communicates with the Blynk IoT cloud platform through WiFi connectivity. The battery parameters are transmitted to the mobile dashboard, where users can monitor real-time voltage values, battery health, system mode, fault status, and operational recommendations. The dashboard provides an intelligent interface for remote battery supervision and diagnostics.

The proposed system provides a reliable, scalable, and efficient solution for battery monitoring applications such as electric vehicles, renewable energy storage systems, and portable power systems by combining embedded control, IoT communication, and automated protection techniques.

WorkFlow Diagram



RESULT

The IoT-Based Smart Battery Management System was successfully designed and implemented using ESP32, Wokwi simulation, and the Blynk IoT platform. The system successfully monitored individual cell voltages of a four-cell battery pack and calculated important parameters such as pack voltage, average voltage, and cell imbalance percentage.

The developed system was tested under different battery conditions by varying individual cell voltages. During normal operating conditions, the system displayed healthy battery status with clear fault status. When abnormal conditions such as high imbalance, over-voltage, under-voltage, or sensor faults were introduced, the system successfully detected the fault and activated the protection mechanism by turning OFF the relay and activating the buzzer.

The Blynk IoT dashboard successfully displayed real-time battery parameters including cell voltages, pack voltage, battery health, fault status, and system mode. The LCD display also provided local monitoring of battery conditions. The implementation of cloud monitoring and automated protection improved the reliability and safety of the battery management system.

The project demonstrates that the proposed IoT-based Battery Management System provides an efficient, reliable, and scalable solution for real-time battery monitoring and protection applications such as electric vehicles, renewable energy storage systems, and portable

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